Kai Xiao

Chinese

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Education

2023 - now **PhD in mathematics**, *Université Claude Bernard Lyon 1 – Institut Camille Jordan*

Thesis : *Sobolev spaces to manifolds: traces, singularities, weak density*
Thesis advisor : Professor Petru Mironescu

2020 - 2023 **Master in mathematics**, *Beijing Normal University*, School of Mathematical Sciences

Thesis : *Regularity of the Solution and Free Boundary of Porous Medium Equations on Riemannian Manifolds*
Thesis advisors : Professor Jingang Xiong

2016 - 2020 **Bachelor in mathematics**, *Beijing Normal University*, School of Mathematical Sciences

Publications

[1] Antoine Detaille and Kai Xiao, *Low regularity maps on Lipschitz manifolds: lifting, pullback of forms, and approximation* (2026), submitted for publication, available at https://hal.science/hal-05776198.

[2] Tristan Bullion-Gauthier and Kai Xiao, *L^2−regularity of minimizers of anisotropic interaction functionals* (2026), submitted for publication, available at https://hal.science/hal-05743076.

[3] Antoine Detaille, Petru Mironescu, and Kai Xiao, *Pullback of closed forms by low regularity maps to manifolds, and applications*, Calc. Var. Partial Differential Equations **65** (2026), no. 198.

Awards and grants

2019 **First Prize (Meritorious Winner) of MCM/ICM**

2018 **Second Prize of the Contemporary Undergraduate Mathematical Contest in Modeling**

Talks

June 2026	**Będlewo**, Calculus of Variations in Material Sciences, *Sobolev maps to manifolds: the distributional Jacobian and its applications*

Teaching experience

2022 - 2023	**Teaching assistant of Graduate Course: Real Analysis**, *Beijing Normal University*
2022	**Teaching assistant of Graduate Course: Partial Differential Equations**, *Beijing Normal University*

Languages

Mandarin	Native speaker
English	Upper intermediate (B2)
French	Utilisateur débutant (A2)

Leisure

Sport	Tennis, basketball
Miscellaneous	Video games, animation